the reverse lever is free to operate the other side; and the locomotive can be run in on one cylinder.

If the crosshead arm, lap and lead lever connector, or lap and lead lever is broken, the same method may be followed. Of course, such of the broken parts as would in any way interfere with running the locomotive would have to be removed.

In the case of *outside* admission valves, the same course may be followed as illustrated in figure 24 and described above; except that the lap and lead lever must be taken down. If this was left up, the radius rod would strike it as the latter moves back and forth to the motion of the link.

MAIN ROD UP, VALVE BLOCKED

In such a case, if the valve has *inside* admission, the locomotive might be blocked as shown in figure 25. For example: suppose the lap and lead lever, lap and lead lever connector, or crosshead arm was broken. Disconnect the radius rod from the lap and lead lever. Suspend it clear of the latter. Tie the lower end of the lap and lead lever ahead to clear the crosshead on the forward stroke. The locomotive can then be run in on one side.

If the valves have *outside*-admission: under conditions similar to those assumed in the case of figure 25, the lap and lead lever must be taken down. In other respects, the same method may be followed.

When the radius rod is suspended, a chain or wire should be used. A rope would be quickly cut through by the motion of the radius rod.

SETTING THE WALSCHAERT VALVE GEAR

If the gear is properly designed, the setting of the Walschaert valve motion is very simple.

It is essential that the length of the eccentric crank, the lengths of the arms of the lap and lead lever and the length of the lap and lead lever connector should check exactly with the drawings.

No change from the drawings in any of the above parts should be made.

Assuming that all the parts check with the drawings, any adjustments which may be necessary to square the motion should be made in the lengths of the eccentric rod, radius bar or valve stem.

Diagrams "A" and "B", figure 26, illustrate respectively the effects on the position of the valve resulting from changes in the lengths of the radius rod and eccentric rod.

It is apparent from diagram "A" that any change in the length of the radius rod "R" will have a corresponding and practically equal result on the position of the valve. That is, if the radius rod is lengthened 1/8 inch the valve will be moved forward on its seat a like amount, or vice versa. Therefore, to move the valve ahead or back any amount, lengthen or shorten the radius bar respectively a like amount.

Reference to diagram "B", figure 26, shows that the effect of a change in the length of the eccentric rod on the position of the valve depends on whether the link block is in the top or bottom of the link.

FIG. 26